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Sensor system for a light fixture having ultraviolet sterilization functionality

Abstract

A sensor system useable in conjunction with one or more environmental fixtures is disclosed. Each of the fixtures preferably includes illumination functionality as well as UV sterilization functionality provided by a fan and UV LED chips in the fixture. One of the fixtures is preferably programmed as a master which executes a control algorithm to control and/or monitor the system. The sensor module includes a plurality of sensors for sensing different environmental conditions where the system is utilized. These sensed conditions are provided to the master fixture, whose control algorithm can use the sensed conditions to control one or more functions in each of the fixtures, such as illumination, UV sterilization, and/or fan speed. The master fixture can output necessary controls to other secondary fixtures in the environment. The system may further communicate with or include external devices that can wirelessly communicate with the system using an application.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. Non-Provisional patent application Ser. No. 17/317,656, filed May 11, 2021, to which priority is claimed, and which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) This application relates to a lighting fixture having the ability to sterilize pathogens such as bacteria, fungi, and viruses.

INTRODUCTION

(2) Lighting fixtures are common in commercial buildings and homes. For example, fluorescent bulb fixtures have been used in commercial buildings and homes for years. Over the last decade or so, Light Emitting Diode (LED) based lighting fixtures have been developed which generally have the same size, shape, and mounting hardware as do traditional fluorescent bulb fixtures (typically 2×2 feet or 2×4 feet). This allows older fluorescent bulb fixtures to be easily replaced by LED based fixtures, which is beneficial because LED fixtures are more energy efficient, more reliable, and easier to maintain when compared with fluorescent fixtures.

(3) Another benefit of LED fixtures is that they can provide radiation suitable to provide disinfection as well as providing visual white light. For example, U.S. Patent Application Publication 2018/0147417 discloses a LED chip useable in a lighting fixture. The LED chip includes a first LED that emits light at 405 nm in the near ultraviolet (UV) range. (The wavelength of light in the visible spectrum ranges from 380 nm at the UV end of the spectrum to 740 nm at the infrared (IR) end of the spectrum). The LED chip also includes a second LED that emits at 450 nm in the blue range of visible light. The LEDs in the chip are coated by a phosphor material, and for the most part the 405 nm radiation passes through the phosphor without absorption. The 450 nm radiation by contrast interacts with the phosphor where it is converted to higher wavelengths, which results in a broader white light emission spectrum. In sum, the LED chip produces an overall spectrum with a peak at 405 nm, as well as a broader-wavelength white spectrum. The inclusion of a significant amount of 405 nm light in the overall spectrum is beneficial, because radiation at that wavelength is known to disrupt certain microbial biological processes. For example, the '417 Publication explains that 405 nm radiation causes reactive oxygen species generation in cells, which in turn prevent cell metabolism and effectively suppresses bacterial growth. 405 nm radiation has also been reported as providing disinfection against fungi. See R. M. Tomb et al., "New Proof-of-Concept in Viral Inactivation: Virucidal Efficacy of 405 nm Light Against Feline Calicivirus as a Model for Norovirus Decontamination," Food & Environmental Virology, Vol. 9(2), pp. 159-67 (2017).

(4) While LED fixture having disinfection properties such as those just described are beneficial, the inventor sees room for improvement. For one, other wavelengths—such as 470 nm—have also

been shown to have antibacterial as well as antifungal properties. See, e.g., A. J. DeLucca et al., "Blue Light (470 nm) Effectively Inhibits Bacterial and Fungal Growth," *Letters in Applied Biology*, Vol 55., pp. 460-66 (2012). But wavelengths such as 405 nm and 470 nm may not be effective against viruses. The article by R. M. Tomb, cited above, investigates the use of 405 nm radiation to inactivate viruses, and while promising results were shown, it appears that much higher doses of 405 nm radiation may be necessary to provide viral disinfection. As 405 nm radiation may be irritating to human eyes, see '417 Publication, it may not be useful to increase the intensity of 405 nm radiation in an otherwise white-light LED fixture in the hopes that it will also kill viruses. (5) Furthermore, the flux or energy density of pathogen-inactivating radiation, such as at 405 and 470 nm, provided by a light fixture may not be sufficient to inactivate air borne pathogens. In short, the volume of the room in which a light fixture is placed may be too large to effectively inactivate air borne pathogens. (6) The inventor discloses a comprehensive solution in the form of a white light LED fixture with effective disinfection properties against bacteria, fungi, and viruses. As discussed further below, the white light LED fixture includes a fan to continuously draw air into the light fixture. The air drawn in is irradiated with UV radiation within the fixture, such as is provided from UV LED chips. The relatively small volume of the light fixture allows the flux or energy density of the UV radiation to be made more intense. After the air is sterilized, it can be put back into the room or building in which the fixture is placed. The white light provided by white light LEDs in the fixture provides illumination, and can further provide significant emission peaks at 405 nm and 470 nm which is also useful to pathogen inactivation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows various views of an improved lighting fixture, having a light box, a fan to draw in air, and a UV sterilization box through which the drawn air passes.
- (2) FIGS. 2A and 2B show white LED chips that can be used in the light box, which preferably produce a white light spectrum having significant near-UV peak wavelengths at 405 and 470 nm, which have shown to be useful to inactivate bacteria and fungi.
- (3) FIG. 3A shows a cross section of the lighting fixture, and FIG. 3B shows the fixture's back plane, diffuser, and one of its circuit boards of the light box.
- (4) FIG. 4 shows a top down view of the UV sterilization box with its cover removed, including UV LED chips and baffles to define a non-linear path for the air drawn into the fixture by the fan.
- (5) FIGS. 5A and 5B shows the system electronics for the fixture, including the provision of power to the driver circuitries for the white LED chips, the UV LED chips, and the fan.
- (6) FIG. 6A shows that sterilized air output from the UV sterilization box can be output back into a room through ports provided in the light box.
- (7) FIG. 6B shows that sterilized air outputs from the UV sterilization box can be combined, and FIG. 6C shows how that sterilized air can be output into the air handling system of a building or house.
- (8) FIG. 7 shows a system of fixtures in an environment, which includes a sensor module for sensing environmental conditions and for allowing control of operation of the fixtures.
- (9) FIG. 8 shows details of the sensor module.
- (10) FIG. 9 shows components on circuit board within the sensor module and each of the fixtures.
- (11) FIG. 10 shows an example of various manner by which the fixtures can be controlled using a control algorithm.
- (12) FIG. 11 shows example of how various sensed environmental conditions can be used to control various functions of the fixtures (e.g., illumination, UV, and fan speed).

(13) FIG. 12 shows a graphical user interface on an application executable on a wireless device for controlling and monitoring the system.

(14) FIG. 13 shows an alternative for the fixture which provides only for UV sterilization and air circulation, but not illumination.

DETAILED DESCRIPTION

(15) An example of a disinfecting light fixture **10** is shown in FIG. 1 in perspective, top down, and bottom up views. The fixture **10** has two main sections: a light box **12**, and a UV sterilization box **14**. Note that these “boxes” **12** and **14** do not need to be box-shaped as shown, and boxes **12** and **14** can instead be understood as any compartment, region, or volume in the fixture **10** however shaped and sized.

(16) The light box **12** includes white LED chips **28** which provide for illumination and whose spectrum additionally and preferably includes significant radiation at 405 nm and 470 nm, as explained further below. The light box **12** includes a fan **20** protected by a grate **22**. The fan **20** is used to draw air into the UV sterilization box **14** where the air is disinfected with UV radiation provided by UV LED chips **82** (FIG. 4), again discussed further below. One or more holes **66** (FIG. 3A) are present in the UV sterilization box **14**, and hose connectors **16a** and **16b** can be fitted in these holes. The air drawn into the UV sterilization box **14** by the fan **20** exits the fixture **10** through these hose connectors **16a** and **16b**, thus outputting sterilized air.

(17) Notice then that the disinfecting light fixture **10** includes different means of providing sterilization of pathogens. The white LED chips **28**, as well as providing white light for illumination, include significant radiation at 405 and 470 nm, which are useful in inactivating at least bacteria and fungi in the air and on surfaces in the room being illuminated, as discussed above. Other air borne pathogens—in particular viruses—are drawn into the fixture by the fan **20** and subjected to high intensity UV radiation provided by the UV LED chips **82** in the UV sterilization box **14**. Such UV radiation should inactivate such air borne viruses, see C. D. Lytle et al., “Predicted Inactivation of Viruses of Relevance to BioDefense by Solar Radiation,” J. Virology (Vol. 79 (22), pp. 14244-52 (2005), and would be expected to provide further sterilization of other air borne pathogens (bacteria and fungi) as well. The air as sterilized by the fixture **10** can then be put back into the room where the fixture **10** is located, or otherwise may be input into the air handling system of the building, as explained further below. Notice that the fixture **10**'s sterilization properties makes it particularly well suited for use in locations where pathogens can be problematic, such as hospitals, nursing homes, etc. Fixture **10** is also useful when incorporated into grow light systems use to grow plants, such as in the system described in U.S. Pat. No. 10,440,900, which is incorporated herein by reference in its entirety. Sterilization is important in this context as well, because growing plants are susceptible to pathogens such as viruses, bacteria, and fungi.

(18) FIG. 1 shows an example of a 2×2 feet (X1) fixture **10**, although the fixture could be made of any shape and size. The UV sterilization box **14** may be smaller in area, e.g., approximately 1.5×1.5 feet (X2). The total height H of the fixture **10** is preferably about six inches, with the light box **12** having a height of about 1.5 inches (H1) and the UV sterilization box **14** having a height of about 4.5 inches (H2). These dimensions are merely one example, and both the light box **12** and the UV sterilization box **14** can have other dimensions as well. The fixture **10** so sized comprises a suitable replacement for traditional fluorescent bulb fixtures. Means for mounting the fixture **10** (e.g., to a room's ceiling) are not shown, but can be of conventional design.

(19) The top view shows that the UV sterilization box **14** can include a section **15** for necessary system electronics, as described later. The bottom view shows the underside of the fixture **10** that which would provide illumination into the room. The fixture **10**'s diffuser **40** (FIGS. 3A and 3B) is removed for easier viewing of underlying structures. Visible from this view are one or more circuit boards **24** which support LED strips **26**. Each LED strip **26** includes a number of white LED chips **28**, which are described in detail with respect to FIGS. 2A and 2B. The size, number, and location of the LED strips **26** is variable, as are the number, type, and location of the white LEDs chips **28**

on these strips. In the example shown, there are four circuit boards **24**, each being approximately 1×1 foot, although a single circuit board **24** could be used as well. Although not yet shown in the figures, the circuit board(s) include a hole **25** to accommodate the fan **20**.

(20) FIG. 2A shows an example of the white LED chip **28**, while FIG. 2B shows the spectrum that results from use of this chip. A white LED chip **28** is shown in top down and cross sectional views, and includes two LEDs **34a** and **34b** mounted to a substrate **30**. A cavity wall **32** surrounds the LEDs **34a** and **34b** and helps to direct light out of the chip **28**. Preferably, the LEDs **34a** and **34b** are different, and emit at different peak wavelengths. For example, LED **34a** can emit at a peak wavelength of 405 nm, while LED **34b** can emit at a peak wavelength of 470 nm. In this example, the LEDs **34a** and **34b** are covered with a phosphor **36**. The phosphor **36**, as one skilled in the art will appreciate, can comprise a mixture of different photosensitive chemicals. Although electrical connections to the LEDs **34a** and **34b** within the chip **28** are not shown, the LEDs can be driven with a current in series or in parallel, or each can be independently driven by their own currents. Each of the LEDs **34a** and **34b** could also be covered with their own unique phosphors as well, or covered with no phosphor at all, although this isn't shown.

(21) As shown in the emission spectrum of the white LED chip **28** in FIG. 2B, it is assumed that the 405 nm radiation largely breaks through the phosphor **36** without being absorbed, and thus this radiation does not substantially contribute to the production of longer wavelengths which would broaden the spectrum. Thus, the spectrum shows a sharp leak at 405 nm. The 470 nm radiation by contrast is designed to interact with the phosphor **36** to produce longer wavelengths, which broadens the spectrum from about 470 to 775 nm, which in sum produces white light useful for illumination. Some amount of the 470 nm radiation is not absorbed by the phosphor **36**, and thus the spectrum includes another peak at this wavelength. Thus, the overall spectrum thus has significant high intensity peaks at 405 nm and 470 nm, but also a broad spectrum that in sum produces white light. In short, the white LED chips **28** in the light box **12** produce white light having significant intensities at 405 and 470 nm. As noted above, inclusion of these peak wavelengths is preferred in the disinfecting light fixture **10** because such radiation impedes (at least) bacterial and fungal growth. One skilled in the art will understand that the disinfection benefits provided by the LEDs **34a** and **34b** are still had even if the peak wavelengths produced by those LEDs are not exactly at 405 nm and 470 nm. In this regard, the LEDs **34a** and **34b** may produce radiation at approximately 405 nm and/or 470 nm, where approximately means a wavelengths that is plus or minus 10 nm from these ideal wavelengths—i.e., from 395 nm 415 nm (in the case of the 405 nm LED **34a**), and from 460 nm to 480 nm (in the case of the 470 nm LED **34b**).

(22) Further sterilization—in particular, of viruses—is provided by the UV sterilization box **14**, although before discussing such details, the construction of the light fixture **10** is described, starting with FIGS. 3A and 3B. The light box **12** includes a diffuser **40**, one or more circuit board sections **24** containing the LED strips **26** as already described, and a back plane **50**. The diffuser **40**, the circuit board(s) **24**, and the back plane **50** are essentially formed in parallel planes inside the light box **12**, and are held in place using a frame **36**. This method of construction is described in U.S. Pat. No. 10,440,900, which was incorporated above. As explained in the '900 patent, the frame **36** can comprise four panels (for each of the four edges of the light box **12**). These panels of frame **36** can be positioned around the diffuser **40**, the circuit board(s) **24**, and the back plane **50**, and then connected to hold these structures securely in place.

(23) The diffuser **40** is positioned between the white LED chips **28** and the room to be illuminated, and is shown in further detail in FIG. 3B. The diffuser **40** operates to scatter light produced by the white LED chips **28** to produce a combined emission spectrum (from white LED chips **28**; FIGS. 2A and 2B) in the illuminated room that is more spatially homogenous. Preferably, the diffuser **40** includes a lens material **43** that is substantially transparent to the emission spectrum that the white LED chips **28** produce. The lens material **43** is typically made of various glass or plastic materials,

such as a polycarbonate light-diffusing white material, and preferably allows good transmissivity of radiation at 405 and 470 nm in particular. The diffuser **40** can further include a brace **41**, preferably made of a metallic material. The brace **41** acts to hold the lens material **43** and the fan grate **22**, as well as providing a decorative element to the fixture **10**. As shown, in this example, the brace **41** divides the diffuser **40** into quarters, and thus the lens material **43** may similarly be formed in quartered segments. Notice that the lens material **43** does not block the fan grate **22**, and thus there is a hole in the lens material to allow for air flow into the light box **12** as promoted by operation of the fan **20**. Note that the fan grate **22** need not be connected to the diffuser **40**. In another example, the fan grate **22** can be connected to the fan **20** or to other structures in the light box **12**, with the fan grate **22** then positioned in the hole in the diffuser **40** during construction.

(24) FIGS. **3A** and **3B** also show details of the back plane **50**. The back plane **50** is preferably formed of a single sheet of a metallic material such as steel or aluminum. The back plane **50** preferably includes a landing **58** to which the motor **44** of the fan **20** can be mounted. As best seen in FIG. **3B**, the landing **58** is connected to the bulk of the back plane **50** via straps **59**, thus defining holes **56** around the periphery of the landing **58**. These holes **56** allow air flow to pass from the light box **12** into the UV sterilization box **14**, as described subsequently. Port **89** in the back plane **50** allows for system signaling to be passed from the UV sterilization box **14** to the electronics on circuit board(s) **24** and to the fan **20**, as discussed later.

(25) As noted, the circuit board **24** can be formed in segments, and FIG. **3B** shows one such segment. Notice that the circuit board segment **24** includes a cut out **25**, which defines a hole when all circuit board segments are positioned in place in the fixture **10**. Again, this hole **25** allows for air flow produced by the fan **20**. Although such details aren't shown, the circuit board(s) **24** are preferably affixed to the back plane **50**, and this can occur in different ways. The circuit board(s) **24** can be screwed to the back plane **50**, possibly using stand offs which provide an air gap between the circuit board(s) **24** and the back plane **50**. Alternatively, and to promote heat conduction away from the circuit board(s) **24**, the circuit board(s) **24** can be affixed in good thermal contact with the back plane **50** using heat conductive tape, paste, or epoxy for example. Although not shown, the outside of the back plane **50** can include heat sinks, as explained in the above-incorporated '900 patent. Note that a benefit of incorporating fan **20** into the light box section **12** is that it promotes heat transfer away from the circuit board(s) **24**, as well as air sterilization functionality.

(26) To summarize, when the fan **20** is operating, air is drawn through fan grate **22**, through the hole **25** in the circuit board(s) **24**, and through holes **56** in the back plane **50** and into the UV sterilization box **14**, whose construction is discussed next. As best shown in FIG. **3A**, the UV sterilization box **14** includes a bottom surface **60**, side surfaces **64**, and a top cover **62**. The inside of the UV sterilization box **14** includes baffles **70** which direct the air flow in a non-linear path and ultimately to holes **66** formed in the side surfaces **64**. As noted earlier, hose connectors **16a** and **16b** are connected to these holes **66**. As will be explained in further detail later, these baffles **70** include UV LED chips **82** to irradiate the air flow as it follows this non-linear path, which is described subsequently with respect to FIG. **4**. The baffles **70** preferably comprise a metallic material, and are preferably affixed to the bottom surface **60**. For example, the bottom edges of the baffles **70** can be bent **72** and affixed to the bottom surface **60** by spot welding, the use of screws, or the use of adhesives. The side surfaces **64** may be similarly attached to the bottom surface **60**. In another example not shown, the baffles **70** may be integrated as a single piece, which can then be dropped into the UV sterilization box **14** during its assembly and affixed in place as necessary.

(27) Components of the fixture **10** may be coated with antimicrobial or reflective materials. For example, the interior surfaces of the UV sterilization box **14** may be coated with Titanium Dioxide. As well as having antimicrobial properties, Titanium Dioxide is highly reflective, thus encouraging reflection of the UV radiation within the UV sterilization box **14**. This is preferred to absorption of the UV radiation, because absorption removes useful energy that could otherwise be used for disinfection of pathogens. In one example, the coating can comprise Paint Shield®, manufactured

by Sherwin Williams. Such a coating can be applied to the vertical surfaces of the baffles **70**, and could also be applied to the underside of the top cover **62**, and the top side of the bottom surface **60**.

(28) The top cover **62** is preferably affixed to the side surfaces **64** using screws **18**. This allows the top cover **62** to be removed to perform maintenance on the fixture **10**, such as to clean or remove the baffles **70** or to repair or replace system electronics, as explained subsequently. The top cover **62** can be affixed to the UV sterilization box **14** using other methods which allow it to be opened and reclosed for maintenance purposes. Although not shown, the hose connectors **16a** and **16b** may also connect to one or more holes provided in the top cover **62**.

(29) The UV sterilization box **14** preferably includes a safety switch **103** designed to cut power to the UV LED chips **82** when the top cover **26** is removed. This is to prevent accidental UV exposure to persons who may be assembling or maintaining the light fixture **10**. This switch **103** can be provided in the UV sterilization box **14** in different ways, but as shown the switch is mounted to the top flange of the side surface **64**. As one skilled will understand, switch **103** includes a contact surface that will be depressed by the top cover **62** when it is connected to the UV sterilization box **14**, thus closing the switch **103** and enabling the UV LED chips **82** to receive power. When the top cover **62** is removed, the contact surface is not depressed and switch **103** is thus opened to prevent activation of the UV LED chips **82**. Operation of the safety switch **103** is discussed further below with reference to FIG. 5A.

(30) The UV sterilization box **14** is preferably fully constructed and then affixed to the light box **12**. In the example shown, this occurs using screws **52** which affix the bottom surface **60** of the UV sterilization box **14** to the back plane **50** of the light box **12**. However, the UV sterilization box **14** and light box **12** can be affixed using different means. Furthermore, the UV sterilization box **14** and light box **12** need not be separately constructed and then attached to each other. Instead, the fixture **10** may be constructed in a manner that integrates the functionality of the UV sterilization box **14** and the light box **12**. Having said this, it can be preferable to manufacture each separately, as this makes it easier to retrofit otherwise standard light boxes **12** with a UV sterilization box **14**.

(31) As best seen in FIGS. 3A and 4, the bottom surface **60** of the UV sterilization box **14** has a hole **61** of preferably the same diameter as the hole(s) **56** formed in the back plane **50** of the light box **12**, which promotes air flow from the fan **20** into the UV sterilization box **14**. Once such air enters the UV sterilization box **14**, it is directed through a non-linear path as directed by the positioning of the baffles **70**. This is best shown in FIG. 4, which shows a top down view of the UV sterilization box **14** with the top cover **60** removed. As shown, the baffles **70** are positioned such that the air flow largely follows a serpentine path from the hole **61** in the bottom surface **60** to the holes **66** in the side surfaces **64** that meet with the hose connectors **16a** and **16b**. The particular manner in which the baffles **70** are positioned in FIG. 4 splits the air flow into four paths. Two of these air flow paths are shown to the right in FIG. 4, although it should be understood that two other air flow paths would be present in the left of FIG. 4, although these aren't shown for simplicity. Note that the air flow paths may not follow a strict serpentine path. For example, the baffles **70** can be positioned to create vortices **74** in the air flow paths. This effectively elongates the air flow path, which exposes air to UV radiation for a longer time, as explained further below. Baffles **72** can be positioned so as to close the air flow paths as necessary to form vortices **74**, as well as to direct the air flow into the baffle structure. Note that the two air flow paths shown to the right eventually join at hole **66** to which hose connector **16b** is affixed. The other two air flow paths on the left join at hole **66** to which hose connector **16a** is affixed.

(32) To more completely sterilize the air in the air flow paths, the non-linear air flow path includes UV LED chips **82**, which may be formed on LED strips **80**. The UV LED chips **82** and strips **80** are shown to the left in FIG. 4, although it should be understood that UV LED chips **82** and strips **80** would also be present in the right of FIG. 4, although this isn't shown for simplicity. In the example shown, the LED strips **80** are affixed to the vertical surfaces of the baffles **70**, as shown in

the plan view at the bottom right in FIG. 4. In this example, there are two UV LED strips **80** spaced vertically on the walls of the baffles **70**, which improves exposure of the air to UV radiation.

(33) Preferably, as much of the non-linear air flow paths are exposed to UV radiation as possible, and so in FIG. 4 the UV LED strips **80** are essentially positioned along the entirety of the lengths of the air flow paths, and further preferably are positioned along at least half of these lengths. The width *d* of the air flow paths around the baffles **70** can may be approximately 1 to 1.5 inches. Assuming that the UV sterilization box **14** is approximately 1.5×1.5 feet (X2, FIG. 1), the length of each of the four air flow paths is approximately 60 to 100 inches, and thus irradiation preferably occurs for at least approximately 30 to 50 inches along these paths. Because the UV radiation may be harmful to people, it is preferable that the UV LED strips **80** not appear in positions where the UV radiation could shine or leak out of the UV sterilization box **14**. Thus, for example, the UV LED strips **80** are not proximate the air input hole **61**, nor are they proximate the output holes **66** to which the hose connectors **16a** and **16b** are affixed. UV LED strips **80** may as shown be placed on both sides of the baffles **70**, which irradiates the air flow paths from opposing sides. While it is preferred to place the UV LED strips **80** on the vertical surfaces of the baffles **70**, they could be placed elsewhere as well, such as on the top side of the bottom surface **60**, or the underside of the top cover **62**.

(34) Assuming that the height of the UV sterilization box **14** is about 4.5 inches (H2, FIG. 1), the total volume of each of the four air flow paths is approximately 360 cubic inches. Fan **20** may for example comprise Part No. 09225VA-12K-AA-cc, manufactured by NMB Technologies Corp., which moves air with a flow rate of 54 cubic feet/minute, which would move air through each of the four air flow paths in parallel at a flow rate of 13.5 cubic feet/minute, or 389 cubic inches/second. As such, each unit volume of air in each flow path is constantly UV irradiated for approximately one second (**360/389**), and with a high flux or energy density because the air is being irradiated almost continuously along the length of each air flow path. Note this is advantageous when compared with other air purification system that use UV radiation to purify air. Typically such systems involve a point UV source which the air to be sterilized rushes passed, meaning that each unit volume of air is only radiated for a short time, which may result in incomplete inactivation of pathogens. By contrast, the air is constantly irradiated in the UV sterilization box **14** along the non-linear paths for an extended period of time, and with a high flux or energy density, thus ensuring more complete disinfection. Of course, the extent to which air is UV irradiated could be varied by changing the flow rate of the fan **20**, changing the length or volume of the air flow paths, changing the intensity and number of UV LED chips **82** used, etc.

(35) In one example, each of the UV LED chips **82** on UV LED strips **80** produces UV radiation with a peak wavelength in the range of 200 to 280 nm, which generally corresponds to the range of UV-C wavelengths. More preferably, the UV radiation has a peak wavelength in the range of 240 to 260 nm, or in the range of 260 to 280 nm. UV radiation in this range has been shown to be particularly useful to inactivate viruses by targeting their nucleic acids. See K. Bergmann, "UV-C Irradiation: A New Viral Inactivation Method for Biopharmaceuticals," America Pharmaceutical Review, Vol 17(6) (November 2014).

(36) While FIG. 4 shows four air flow paths each following a non-linear path, and two output holes **66**, it should be understood that this is just one example. There could be more or less air flow paths established with the UV sterilization box **14**, or more or less holes **66**. For example, a single non-linear path could comprise a spiral in which air input via hole **61** spirals around the box **14** at increasing diameters, until the sterilized air eventually exits the box at a single output hole **66**.

(37) FIG. 4 shows further options that can be included with the UV sterilization box **14**, and in particular with the hose connectors **16a** and **16b**. As shown in the upper right, the interior diameter of the hose connectors **16a/b** includes a one-way valve **93** that only allows sterilized air to pass out of the UV sterilization box **14**. The hose connectors **16a/b** may also include a pressure relief valve **95** which is designed to vent the sterilized air should it exceed the valve **95**'s pressure. The interior

diameter of the hose connectors **16a/b** can also include filters **97**, such as charcoal filters, to further filter particulates and pathogens, and to also work as an anti-odorant. The anti-odorant properties of the filter **97** can be particularly useful when the fixture **10** is used in a grow farm setting and when the plants being grown have strong odors (e.g., *cannabis*). The filters and valves need not necessarily be positioned within the hose connectors **16a/16b**, but could comprise discrete components connected to the hose connectors **16a/b** outside the box **14**. Although not shown, the air flow paths within the UV sterilization box **14** could include filters and valves at various points as well.

(38) As shown in FIG. 4, the UV sterilization box **14** can include an electronics section **15**. This section **15** can be walled off from the baffles **70** and the air flow paths by a wall **90**. Section **15** can include the driver circuitry **92a** for driving drive the white LED chips **28** in the light box **12** and driver circuitry **92b** for driving the UV LED chips **82** in the UV sterilization box **14**. It is preferable that the driver circuits **92a** and **92b** be separate because the white LED chips **28** and UV LED chips **82** may have different driving requirements (voltages, currents, power, etc.). Driver circuitries **92a** and **92b** could also be integrated in another example.

(39) Electronics section **15** can include or more ports **86** which receive AC power **100** (FIG. 5) from outside the fixture **10**, e.g., from a socket or other power source or line to which the fixture **10** is connected. The section **15** may also include a port **88** in the bottom surface **60** to allow signaling to be output from driver circuitry **92a** to the white LED chips **28** in the light box **12**. Port **88** can correspond in position to a similar port **89** in the back plane **50** of the light box **12** (see FIG. 3B). Although not shown, one skilled will understand that such signaling will connect to connectors or contacts on one or more of the circuit board(s) **24**. AC power for the fan **20** can also pass through the ports **88/89**.

(40) Electronics section **15** may also include one or more ports **84** to allow signaling to be output from driver circuitry **92b** to the UV LED chips **82** in the UV sterilization box **14** and to the safety switch **103**. One skilled will understand that such signaling will connect to each of the UV LED strips **80**. In this regard, it can be useful to connect the various UV LED strips **80** within the UV sterilization box in a manner to reduce the amount of signaling and connections required. Although not shown, the bottom surface **60** can include a circuit board to assist in routing signaling to the UV LED strips **80**. This circuit board may include, or be in addition to, a circuit board **11** that is used to control the fixture, as explained further later. Preferably, port(s) **84** are optically blocked after the signaling has passed through to prevent UV light from entering electronics section **15**. It is preferable to include the system electronics within section **15** so it can be easily accessed. For example, top cover **62** of the UV sterilization box **14** can be removed (using screws **18**, FIG. 3A), thus allowing access as necessary to maintain or replace system electronics. System electronics could also be located in the light box **12**. The size of electronic section **15** can vary depending on the size of the system electronics that are supported.

(41) System electronics are shown in a first example in FIG. 5A. AC power provides a voltage V_{ac} , which is provided to the white LED driver circuitry **92a**, to the UV LED driver circuitry **92b**, and to the fan **20**. Although not shown, it should be understood that V_{ac} may be processed (transformed, rectified to DC voltages, etc.) prior to being provided to the driver circuitries **92a** and **92b** and fan **20** in accordance with their input power needs. White LED driver circuitry **92a** typically provides a compliance voltage V_w as necessary to provide a current I_w necessary to drive the white LED chips **28**. A regulator **94a** can be used to control I_w , as is well known. UV LED driver circuitry **92b** is similar, and provides a compliance voltage V_{uv} as necessary to provide a current I_{uv} necessary to drive the UV LED chips **82**, with a regulator **94b** controlling I_{uv} . In one example, the power required by the fixture **10** may comprise about 100 Watts, with the white LED chips **28** requiring about 60 W, the UV LED chips **82** requiring approximately 30 W, and the fan requiring about 10 W.

(42) It may be desired to separately control one or more aspects of the fixture **10**. For example, it

may be desired at a given time to drive only the white LED chips **28** to provide illumination to a room the fixture **10** is placed in, but to not drive the UV LED chips **82** to provide UV disinfection. Conversely, it may be desired at a given time (e.g., at night) to drive only the UV LED chips **82** to provide UV disinfection, but to not drive the white LED chips **28** to provide illumination. In this regard, the fixture **10** can include or be controlled by one or more switches **100**, **102**, or **104**. For example, switch **100** comprises a master switch used to control all operations of the fixture, i.e., to control driving the white and UV LED chips **28** and **82**, and the fan **20**. Switch **102** can be used to independently control the white LED chips **28**. Switch **104** can be used to independently control the UV LED chips **82** and the fan **20**. Switch **104** is useful because it would normally be expected that the fan **20** and UV LED chips **82** would be enabled together, with the fan **20** drawing air flow into the UV sterilization box **14** that includes the chips **82**. That being said, the UV LED chips **82** and fan **20** could also be independently controlled by their own switches. Any of the switches shown could comprise wall-mounted switches to which the fixture **10** is connected. Alternatively, the switches can appear in the light fixture (section **15**) as part of the system electronics. In this respect, the switches may be controlled by a remote control, with system electronics including a wireless receiver (e.g., a Bluetooth receiver) for receiving input from the remote control. For example, application **200**, explained further below, can be used to wirelessly control the switches.

(43) System electronics can further include a safety switch **103**. As described earlier, this switch **103** is designed to open to cut power to the UV LED chips **82** (e.g., via driver circuitry **92b**) when the top cover **62** is removed from the UV sterilization box **14**. As shown, safety switch **103** is in series with switch **104**, and so would also disable power to the fan **20**. However, switch **103** could also be located in the circuitry to cut power to only the LED driver circuitry **92b**.

(44) FIG. 5B illustrates another embodiment of system electronics. The system electronics allows the fixture **10** to be powered using either AC power **100** or using a battery **13**, which would preferably be present in the electronic section **15** of the fixture **10**. The battery **13** may be either a rechargeable battery or a primary battery. Though not shown in FIG. 5B, the system electronics may include charging circuitry for recharging the battery **13** when the fixture **10** is connected to AC power **100**. The illustrated embodiment of the system electronics includes a relay **17**, such as the illustrated single pole double throw (SPDT) relay, which is configured to automatically switch between battery **13** power and AC power **100** depending on whether AC power **100** is available. Other switching configurations will be apparent to a person of skill in the art. In FIG. 5B, the fan **20**, the UV LEDs **82**, and the white LEDs **28** are as described earlier. Driver circuitries **92a** and **92b** (FIG. 5B) are integrated as driver circuitry **91**, which is also used to drive the fan **20**. Note that the live (L) and neutral (N) terminals of the driver circuitry **91** are denoted in FIG. 5B. Various switches described in FIG. 5A are not shown in FIG. 5B for simplicity but could be present.

(45) The system may also be provided a source of solar power **85**. This source **85** may comprise solar panels included in a premise where the fixture **10** is installed, or which optionally is sold with the fixtures **10**. Solar power from source **85** may power the fixtures **10**, and also may be used to charge the batteries **13** in the fixtures. Again, the system may include battery charging circuitry, but this detail isn't shown in FIG. 5B. In this respect, AC power **100** isn't strictly necessary, and instead solar power can operate the fixtures **10** when the sun is present, leaving it to the batteries **13** to provide power when the sun isn't present.

(46) As discussed above, the UV sterilization box **14** includes one or more hose connectors **16a** and **16b** which output sterilized air, and such sterilized air is preferably distributed back into the room or building in which the fixture **10** appears. FIGS. 6A-6C show different examples of how this can occur. Sterilized air can also effectively be disposed with, such as by venting such air into the plenum space in a building or house, or through a vent to the outside environment.

(47) FIG. 6A shows an example in which the sterilized air is output back into the room through the fixture **10** itself. In this example, which shows a larger light fixture (2×4 foot), the light box **12** includes one or more hose ports. Two such ports **110a** and **110b** are shown in FIG. 6A, and may

comprise hose connectors allowing them to be joined to the hose connectors **16a** and **16b** by hoses **112a** and **112b** as shown. The ports **110a** and **110b** in this example proceed through holes in the back plane **50**, the circuit board(s) **24**, and the diffuser **40** of the fixture **10**. Although not shown, the air output from the hose connectors **16a** and **16b** can be combined (e.g., FIG. **6B**) and put back into the room through a single port **110** in the light box **12**, or through more than two ports.

(48) FIG. **6C** shows that the air output from the hose connectors **16a** and **16b** can be placed into the air handling system in a building in which one or more fixtures **10** are placed, thus providing sterilized air to one or more rooms in the building. In this example, it is assumed that the building has a number of rooms (two of which **120a** and **120b** are shown) with each room having a number of fixtures **10** (three in each as shown). The building includes an air handler **118** with an input **126** and an output **128**. One skilled will recognize that the duct work of an air handling system could include other components that are not shown, such as fans, exhaust vents, fresh air inputs, etc. Each room **120a** and **120b** has a supply vent **124** connected to the output **128** and a return vent **122** connected to the input **126**. FIG. **6B** shows that the air output from the hose connectors **16a** and **16b** in a given fixture **10** can be combined (e.g., FIG. **6B**) using a junction **114**, which outputs to an output hose **116**. Junction **114** and output hose **116** could also be fit with filters (**97**) and valves (**93**, **95**), as explained earlier with reference to FIG. **4**. The outputs from several output hoses **116** can be connected as shown in FIG. **6C**, and connected by another hose or duct work to any convenient point in the air handler duct system, including the return line of a given room (**130**), the input **126** to the air handler **118** (**132**), the output of the air handler **118** (**134**), or to the supply line of a given room (**136**). In any of these examples, the sterilized air is ultimately provided back into the room(s).

(49) FIG. **7** shows further details by which one or more fixtures can be integrated and controlled in an environment, and further shows how environmental sensing can be included with the system. In this example, four fixtures **10a-10d** are shown, which can be mounted in the ceiling of a room for example, as explained earlier. Also shown is a sensor module **150** which includes sensors that are used to sense various environmental conditions in the room. Such sensed conditions can be used to control various functions of the fixtures, such as operation of the fan **28**, intensity of the UV LED chips **82s** used for sterilization, and the intensity of the white LED chips **28** used for illumination, as explained further below.

(50) The sensor module **150** can communicate with each of the fixtures **10i**, either by electronic cabling or using a short range RF communication format such as Bluetooth™. In the depicted example, and more preferably, the sensor module **150** communicates with a particular one of the fixtures (**10a**) via communication link **170**. This fixture **10a** can be programmed in the system as a “master” fixture, while other fixtures (**10b-10d**) can be programmed as “secondary” fixtures, as explained further below. The master fixture **10a** can operate a control algorithm **320** to control operation of the system, explained further below. The master fixture **10a** can communicate with the secondary fixtures **10b-10d** using electronic cabling, but again the use of short range RF communication links **180** such as Bluetooth is preferred. Preferably, the master and secondary fixtures are identical in construction and differ only in whether they are programmed as master or secondary fixtures.

(51) The sensor module **150** may also communicate with an alert indicator **190**, which includes one or more indicators such as LEDs **200** that can be used to provide certain visual alerts to persons in the environment indicative of environmental conditions that the sensor module has sensed. For example, the alert indicator **190** may indicate a green light is the sensor module **150** has not detected any unusual environmental conditions or if the system is otherwise working properly. The alert indicator **190** may indicate a red light if dangerous environmental conditions have been detected or if the system is not working properly. The alert indicator **190** may also indicate alerts in other ways. For example, the alert indicator **190** may include a speaker to audibly indicate alerts. A person seeing or hearing an alert at **190** may consult an application **230** to understand the details

concerning the alert, as described further below. Although not shown, the alert indicator **190** may be integrated with the sensor module **150**, with the sensor module **150** providing necessary alerts. However, because the alert indicator **190** may be distracting, it may be beneficial to place it in a non-obtrusive place, such as in the ceiling at a corner of a room. The sensor module **150** by contrast is preferably placed in the center of a ceiling of a room to most evenly sense the environment's conditions. The fixtures **10i** would preferably be placed in the room as necessary to provide illumination and sterilized air, as described above. In the example shown in FIG. 7, the alert indicator **190** is hardwired to the sensor module **150**, but these components could also communicate wirelessly (e.g., by Bluetooth).

(52) Relevant electronics in the sensor module **150** are integrated on one or more circuit boards **151**, while relevant electronics in the fixtures **10i** are integrated on one or more circuit boards **11** preferably located in the electronics section **15** of the fixtures **10i**, as explained earlier. Components present on circuit boards **151** and **11** are explained later with reference to FIG. 9.

(53) As discussed earlier, each fixture **10** may each include a battery **13** for providing power to the fixtures. The output of these batteries **13** may be joined at a common power bus **153**. Further, power bus **153** may be routed to the sensor module **150** to provide power to the sensor module **150**. However, as explained further below, the sensor module **150** may also have its own battery **310**.

(54) Wireless communication in the system of FIG. 7 provides flexibility to allow the system to communicate with other devices to useful ends. For example, the system can communicate with an electronic device **210**. This device **210** may be a general-purpose computing device such as a personal computer, tablet, smart cell phone and the like. Device **210** can run an application (app) **230** to allow a user to both control the system (illumination and/or UV sterilization), and to review important information reported from the system (such as alerts, or output from the sensors in the sensor module **150**). Notice in the depicted example that the device **210** communicates with the master fixture **10a** (again preferably using Bluetooth), although device **210** may also communicate with the sensor module **150** or other intermediaries in the system. The system (via master fixture **10a**) can also communicate with other networks and systems. For example, via a Bluetooth router, the system may ultimately communicate with a Wifi™, cellular or internet network **240**. Master fixture **10a** may also include communication circuitry operable using WiFi™, cellular, or internet formats to allow connection to network **240** directly if necessary. Connectivity via such other networks **240** allows the system to be accessed via a remote device **220**, which may again comprise a personal computer, tablet, smart cell phone and the like, and which may again run app **230** to allow for system control and monitoring.

(55) FIG. 8 shows the sensor module **150** in further detail. The module **150** includes a housing **152**, which may be formed of plastic. The housing **152** may be formed with clips (not shown) allowing it to be secured to a bracket (not shown) affixable to a ceiling—similar to the manner in which a smoke detector is affixed to a ceiling. The control module **150** can also comprise a stand-alone device freely positionable in the environment being monitored. Within the housing **152** is the above-mentioned circuit board **151**, which among other things includes various sensors **330** that monitor various environmental conditions, as explained further below. The housing **152** preferably includes air channels **156** to allow air in the environment to enter the sensor module **150** and come into contact with the sensors **330**. Note that sensors **330** can be positioned on either side of the circuit board **151**. Certain sensors described subsequently may require a line-of-sight to the environment being monitored, and in this regard the housing **152** can include one or more openings **154** to allow such sensors to protrude through the housing and be directly exposed to the environment.

(56) Components present on circuit board **151** in the sensor module **150** are shown in FIGS. 8 and 9. As shown, the circuit board includes telemetry circuitry **304** to enable wireless communications with other devices, such as the master fixture **10a**, and possibly also other devices such as devices **210** and **220** (FIG. 7). In one example, telemetry circuitry **304** comprises a communication antenna

and chip sets designed for communications accordingly to a particular format, such as Bluetooth, WiFi, cellular, etc. In the depicted examples, telemetry circuitry **304** comprises a Bluetooth chip set and antenna. Circuit board **151** also preferably comprises controller circuitry **306** such as a microcontroller. A significant function of the controller circuitry **306** is to receive data being reported from the sensors **330**, and to organize such data for communication, for example to the master fixture **10a**.

(57) Although not shown in FIG. **8**, circuit board **151** in the control module **150** can be connected (plugged in) to a source of AC power (e.g., **100**, FIG. **5**), and circuit board **151** may include one or more rectifier/regulators to produce DC power to power the circuit board **151**'s various components. Additionally, or alternatively, the circuit board **151** can receive power from a battery **310**. In a preferred example, the battery **310** is rechargeable, and is recharged (using AC power **100**) by battery charging circuitry (not shown) on circuit board **151**. Battery **310** may also comprise a permanent battery, requiring the user to replace the battery **310** once it is spent.

(58) Providing a battery **310** in the sensor module **150** is preferred in case there is an interruption of AC power **100**. In the case of a power interruption, the sensors **330** can continue to function, at least for a time (e.g., 90 minutes). Furthermore, battery **310** can provide a safety feature, and can allow the sensor module **150** to operate as an emergency illumination source. In this regard, the circuit board **151** can include one or more light sources, such as LEDs **308**. These LEDs **308** (mounted to the underside of the circuit board **151**) can protrude through openings **158** provided in the housing **152** to provide low-level illumination in the environment in case of a power outage. Further, illumination provided by LEDs **308** can also be controlled by a user of app **230**, or LEDs **308** can illuminate automatically without user intervention, such as at certain times of day (e.g., after hours) or in accordance with information read from the sensors **330**. For example, if the EM sensor **364** (FIG. **9**) senses that the room is dark, perhaps because illumination in the fixtures **10i** is off, controller circuitry **306** can activate the LEDs **308**. LEDs **308** are hereinafter referred to as emergency LEDs **308**, even though they do not necessarily illuminate during emergencies.

(59) The sensor module **150** as mentioned can include a plurality of different sensors **330**, each designed to detect a particular condition of the environment in which the sensor module is placed. For example, and as shown in FIG. **9**, the sensor module **150** can include: a temperature sensor **332**; a relative humidity sensor **334** (e.g., part number HTS221, manufactured by STMicroelectronics. This part also includes a temperature sensor **332**); an atmospheric pressure sensor **336**; a dew point sensor **338**; a particulate sensor **340** (e.g., part number SN-GCJA5, manufactured by Panasonic Photo & Lighting Co., Ltd.); an Air Quality Index (AQI) sensor **342**; a Total Volatile Organic Compound (TVOC) sensor **344** (e.g., part number ZMOD4410, manufactured by Renesas Electronics Corp. This part also includes an AIQ sensor **342** and a CO.sub.2 sensor **350**); an HCHO (formaldehyde) sensor **346**; a carbon monoxide (CO) sensor **348** (e.g., part number SGX-4CO, manufactured by SGX Sensortech); a carbon dioxide (CO.sub.2) sensor **350** (e.g., part number Telaire T6703, manufactured by Telaire Co.); a sulphur dioxide (SO.sub.2) sensor **352**; a nitrous dioxide (NO.sub.2) sensor **354**; an ozone (O.sub.3) sensor **356**; a lead (Pb) sensor **358**; a radon (Rn) sensor **360**; an infra-red (IR) motion sensor **362** (e.g., part number EKMB130911, manufactured by Panasonic, Inc.); an ElectroMagnetic (EM) sensor **364**, which can be used to assess the spectrum of electromagnetic radiation in the environment; and biosensors **366**, which can use to determine whether pathogens are present in the environment, including bacteria, viruses, fungi and the like.

Preferred part numbers for some of these sensors **330** are listed above, although one skilled will recognize that there are many different types of these sensors. It is not required that the sensor module **150** have all of the above-listed sensors: certain of these sensors **300** may only be important in particular environments (e.g., manufacturing environments) and therefore may not be necessary when sensor module is used in different environments (e.g., office environments). The above sensors **330** preferably output values indicative of the quantity of the conditions they are

sensing, although certain sensors may operate in a binary fashion and only output whether or not a particular condition has been sensed. Possible sensors **330** as shown in FIG. **9** are not an exhaustive list, and sensor module **150** can include still other environmental sensors presently existing or developed in the future. In one example, a sensor module **150** can include all possible sensors **330** potential customers might desire, with the manufacturer programming the sensor module **150** (e.g., using app **230**, FIG. **12**) to only enable sensors **330** relevant to a particular customer or environment. As mentioned above, sensors **330** requiring a line of site to the environment can be placed in an opening **154** (FIG. **8**) of the sensor module's housing **152**, such as the IR motion sensor **362**.

(60) FIG. **9** also shows components present on the circuit board **11** for each of the fixtures **10i**. The circuit board **11** includes controller circuitry **300** (e.g., a microcontroller), which is programmed with a control algorithm **320**, explained further below. The controller circuitry **300** receives a master/secondary (M/S) input **303** which informs whether the fixture **10i** in which circuit board **11** is placed is to operate as a master (e.g., **10a**) or a secondary (**10b-10d**) fixture. By programming the M/S input **303** accordingly, a master fixture will be configured to receive information from the sensor module **150**, and to control the secondary fixtures using the control algorithm **320**. By contrast, programming the M/S input **303** to designate a fixture as a secondary fixture will configure that fixture to only communicate with and be controlled by the master fixture **10a**. Thus, the control algorithm **320**, if programmed into the controller circuitry **300** of a secondary fixture, will be disabled. In one example, the M/S input **303** may comprise a dip switch or button which may be set in each fixture **10i** by the manufacturer or the customer. Alternatively, the M/S input **303** may be set wirelessly using devices **210** and **220** via app **230** (FIG. **7**). It is assumed in FIG. **9** that the circuit board **11** has been programmed at M/S input **303** to operate as a master.

(61) Circuit board **11** includes telemetry (e.g., Bluetooth) circuitry **302** capable of receiving information wirelessly transmitted from the sensors **330** in sensor module **150** and communicating with the other fixtures **10i**. Environmental conditions sensed by the sensors **330** are denoted as **S1**, **S2**, **S3**, etc., in FIG. **9**, each representing different reported sensed condition (e.g., temperature, humidity, pressure, dew point, particulates, etc.). Such sensor information is received at control algorithm **320** programmed into controller circuitry **300**. The control algorithm **320** operates to control different aspects of the fixtures **10i** by issuing various control signals, **C**.

(62) In a preferred example, a number of different control signals **C** are issued for each fixture **10i**. For example, control signals **Cai** comprise control signals for fixture **10a**, with **Ca1** controlling the fan **20** (e.g., whether it is on or off, and the speed of the fan), **Ca2** controlling the UV LED chips **82** (e.g., whether they are on or off, and their intensity), and with **Ca3** and **Ca4** controlling the white LED chips **28** (again whether they are on or off, and their intensity). Specifically, **Ca3** can control the LEDs **34a** (e.g., peaking at 405 nm; see FIGS. **2A** & **2B**), while **Ca4** controls LEDs **34b** (e.g., peaking at 405 nm). Because the fixture in FIG. **9** has been programmed to act as a master fixture, notice that these control signals **Cai** can be sent by wiring to the appropriate hardware components in this fixture. Thus, fan **20** receives **Ca1**, UV LED driver circuitry (**92b**) driving UV LED chips **82** receives **Ca2**, and white LED driver circuitry **92a** driving white LED chips **28** (LEDs **34a** and **34b**) receives **Ca3** and **Ca4**. Such driving circuitry is shown on circuit board **11**, but could be present on a different circuit board in the fixture. One skilled will understand that these control signals **Cai** as output from the control algorithm **320** may be processed as necessary to interface with the hardware components that they control.

(63) The control algorithm **320** also determines control signals **C** necessary for each of the secondary fixtures (e.g., **10b-10d**), and uses the telemetry circuitry **302** to wirelessly transmit those control signals to those fixtures. For example, control signals **Cbi** (**Cb1** for the fan, **Cb2** for UV LED chips **82**, and **Ca3** and **Ca4** for white LED chips **28**) are to be transmitted to fixture **10b**, control signals **Cci** are transmitted to fixture **10c**, and so on. Again these control signals may undergo some processing at the receiving fixtures as necessary to interface with their hardware components. Thus,

control of all fixtures **10i** in the system is provided by the control algorithm **320** in the programmed master fixture **10a**. Such control in certain circumstances can occur in accordance with the information S_i that are sensed by the various sensors **330** in the sensor module **150**, as explained further below.

(64) The control algorithm **320** may also receive an indication of the present time and date, designated as 't,' which may be provided from clocking circuitry on the circuit board **11**. Knowing the present time can be useful in the system for a number of different reasons. For example, and as discussed further below, the control algorithm **320** may be programmed take different actions at different times of day. Furthermore, the circuit board **11** may also include a memory **312** which keep a log file **314** logging important events that occur during operation of the system, and entries in the log file **314** may be time stamped with the time/date t. The log file **314** may be reviewable by a user using the app **230** on devices **210** or **220** (FIG. 7), as explained further below. The log file **314** for example may store information S_i reported by the sensors **330**, thus allowing the user to review the current value of these sensor outputs or how they are varying as a function of time. The log file **314** may also store information regarding the various control actions that algorithm **320** has undertaken (e.g., the status of control signals C), which may also be reviewed as a function of time.

(65) The control algorithm **320** may receive a number of inputs to control its operation, in addition to the sensor inputs S_i . For example, the control algorithm **320** may receive one or more programs P_i , which may specify how control algorithm will control the various fixtures **10i** in terms of illumination, UV sterilization functionality, and fan operation. A program may contain information that allows the control signals C to be determined for each hardware component in each fixtures **10i**. A program P_i may also include timing (e.g., a schedule) during which it will operate.

(66) The control algorithm **320** can also operate automatically to control illumination and disinfection in the system dependent on what the sensors **330** sense (S_i). In this regard, the control algorithm **320** can be programmed with a number of parameters X_i , which dictate how the control algorithm **320** will operate. Parameters X_i may comprise thresholds for certain sensed information S_i , or weights that the control algorithm **320** may apply to such sensed information. Parameters X_i may also include variables that allow the control algorithm **320** to make automatic adjustments based on a history of use of the system, and as such the control algorithm **320** may comprise a machine learning algorithm, which parameters X_i comprising inputs to the machine learning algorithm. Programs P_i and parameters X_i may programmed into control algorithm **320** by the system manufacturer, or may also be wirelessly programmed using the application **230** on devices **210** and **220** for example.

(67) FIG. **10** shows a simple example of different ways that the system can be controlled at different points in time. In this example, it is assumed that control varies during a day, but longer periods of time (weeks, months) etc., could also controlled. Using longer periods may be sensible, as needs for system control may vary on weekends or holidays, or in hotter or colder months for example. The example in FIG. **10** assumes that the system is installed in an office conference room, although again this is merely an example as the system can be used in other environments (industrial, greenhouse, etc.) as well.

(68) Two programs **P1** and **P2** are specified for use at different time periods. Program **1** operates at the end of a work day, from 8 pm to midnight in this example. Because people would not be expected to be present during these hours, program **1** specifies that illumination provided by the fixtures (white light LEDs **28**) should be off (or low). Although not shown in FIG. **10**, program **1** may also allow illumination when people are detected in the room, e.g., by motion IR sensor **362** (FIG. **9**). Emergency LEDs **308** on the sensor module **150** may be illuminated to provide low level lighting. It may be reasonable to well sterilize the air in the room at the end of a work day, and program **1** sets fans **20** and UV sterilization (provided by UV LEDs **82**) in the fixtures **10i** to a maximum intensity. During program **1**, sensed environmental conditions are not important to control of the system, and hence may be ignored by the control algorithm **320** (except perhaps the

output from motion IR sensor **362**). However, such environmental conditions can continue to be sensed, and if necessary alerts regarding certain environmental conditions (in particular, those that present dangers, such as CO, O.sub.3, Pb, Rn, etc.) can be generated by the control algorithm **320**. Such alerts can be transmitted to the sensor module **150** and in turn to the alert indicator **190** (FIG. 7), as well as to devices **210** and **220** where the nature of the alert can be viewed in the application **230**.

(69) Program **2** runs from midnight to 6 am. At this point, illumination may be of lesser concern, and so illumination provided by the fixtures **10i** may be programmed off (even if motion is detected by motion IR sensor **362**). Likewise, emergency LEDs **308** in the sensor module **150** may be off. Finally, because the air was earlier well sterilized during program **1**, fans and UV sterilization may be programmed as off during program **2**. As with program **1**, sensed environmental conditions may not be important to control of the system during program **2**, and hence may be ignored by the control algorithm **320**, except to issue alerts.

(70) From 6 am to 8 pm, generally corresponding to work hours, a specific program may not run. Instead, illumination, fans, and UV sterilization may be automatically controlled by the control algorithm **320** in accordance with sensed environmental conditions. (Providing automated control via the control algorithm **320** may also comprise a “program,” e.g., program **3**). It may be sensible to provide automated control during this time period because environmental conditions in the conference room may be changing significantly. As noted above, automatic operation of the control algorithm **320** can in one example be controlled using parameters X_i , which may comprise thresholds for certain sensed information S_i , or weights that the control algorithm **320** may apply to such sensed information.

(71) FIG. **11** generally explains how various sensed environmental conditions can be used by control algorithm **320** to provide control in the system. For example, temperature (T) (sensor **332**) can affect pathogens such as viruses, bacteria, and fungi, because such pathogens are less inhibited at lower temperatures and more inhibited (and perhaps even eradicated) at higher temperatures. As such, it may be reasonable for the system to provide higher intensity UV radiation at lower temperatures (below a temperature threshold X_1), and to provide lower intensity UV radiation at higher temperatures (above X_1).

(72) Alternatively, the sensed temperature can be weighted (along with other sensed environmental conditions) to determine how UV sterilization in the system should be controlled. Weights can be useful as a control mechanism, because they can quantify how significantly a particular sensed environmental condition should be treated relative to the system function being controlled. For example, assume temperature is accorded a low weight X_1 relative to UV sterilization. A different environmental condition, such as the amount of detected Total Volatile Organic Compounds (TVOC) (sensor **344**), may be accorded a higher weight X_2 relative to UV sterilization because higher amounts of UV radiation may break down such compounds. If the sensed temperature is high (suggesting that UV sterilization should be decreased), and the sensed amount of TVOCs is also high (suggest that UV sterilization should be increased), the system may be in conflict as to whether UV sterilization should be decreased or increased. However, the different weights (X_1 and X_2) in this circumstance would suggest that the control algorithm **320** increase UV radiation. In effect, control algorithm **320** gives greater weight to sensed TVOCs over sensed temperature when deciding how to control UV sterilization.

(73) Different sensed environmental conditions may impact the control of different control functions in the system. For example, while temperature is relevant to adjusting the intensity of UV, it may not be relevant to adjustment of the fans or the illumination. By contrast, relative humidity (RH) (sensor **334**) is relevant to controlling the fans **20** in the fixtures: if the RH is low, the speed of the fans **20** can be decreased, but if RH is high, the speed of the fans may need to be increased to promote the flow of moist (heavy) air into the fixtures. Other sensed environmental conditions may be relevant to illumination control. For example, the motion IR sensor **362** may determine the

number of persons present in the conference room, and increase (or turn on) the illumination when one or more persons are detected. Some sensed environmental conditions may be relevant to control of more than one function in the system. For example, the motion IR sensor **362** may also control the fan and UV sterilization, increasing both when the number of persons in the room is high, and decreasing both when the number is low.

(74) Still other sensed environmental conditions may not be used by the control algorithm **320** for system control, but instead may only be recorded and reported, or used to issue alerts. For example, the detection of lead (Pb sensor **358**) may not be relevant to adjustment of illumination, fans, or UV sterilization, and thus may not be used for system control. However, because lead is dangerous, it can trigger an alert (e.g., at alert indicator **190**), and may be logged (**314**, FIG. **9**) for user review at devices **210** or **220**. Still other sensed environment conditions may simply be logged and reported, but not used to set any alarms (e.g., atmospheric pressure, dew point, sensed EM spectra, etc.).

(75) Returning again to FIG. **10**, automatically control by the control algorithm **320** in accordance with sensed environmental conditions generally occurs from 6 am to 8 pm. However, such automated control may be interrupted during a period of manual control, as shown from 2 pm to 4 pm. During this time period (perhaps corresponding to a large meeting in the conference room), a user may have used app **230** on devices **210** or **220** for example to manually control one or more functions in the system. For example, during manual control, the user may have decided to provide maximum illumination, maximum UV sterilization, or maximum fan speed, or none at all, thus overriding what control algorithm **320** would otherwise automatically suggest. During manual control, the user may also program the various fixtures to operate differently. Manual control is represented in FIG. **9** by inputs M_i , received from the devices **210** or **220**, which may correspond to a user selection of the various control signals C for the various fixtures. Note also during manual control that a user may choose to run a particular program for an unscheduled amount of time. It is assumed in FIG. **10** that the user has caused the control algorithm **320** to return to automatic control at the end of the meeting (e.g., at 4 pm), although the control algorithm **320** may also automatically return to its schedule at some default time after manual control was engaged. Lastly, although not shown, note that programmed/manual control and automated control may not be mutually exclusive in the system. For example, control algorithm **320** may provide for automated control of UV sterilization and the fans in the system, leaving illumination functionality subject to manual or programmed control.

(76) An example of a graphical user interface (GUI) for application **230** as may nm on devices **210** or **220** is shown in FIG. **12**. As noted earlier, application **230** can be used by a user to control the system, and to review status information reported by the system. In the example shown in FIG. **12**, each of the system's components is shown in the application, including the various fixtures 10_i , the sensor module **150**, and the alert indicator **190**. Preferably, each of the displayed components can be selected in the GUI to display additional information about the component and/or to allow for data to be input. For example, selecting the master fixture allow for the control algorithm **320** operative in this component to be programmed, for example by allowing the parameters X_i discussed above to be input. Selecting the master fixture also allows the log file **314** to be reviewed. Further, selecting any of the fixtures 10_i (master or secondary) allows the current settings at that fixture (e.g., illumination, fan, UV—e.g., their control signals C) to be reviewed.

(77) Selecting the sensor module **150** provides further options to allow various sensors **330** in the sensor module **150** to be enabled. As noted above, not all sensors **330** provided by sensor module **150** may be necessary to use in a given environment. Selection of the sensor module **150** also allows the current sensor outputs S_i to be reviewed, although this may also be reviewable via the log file **314**. Selecting the sensor module **150** can also provide an option for to allow a user to review the status of the battery **310** in the module, such as how full this battery is and/or how much more operating time is remaining. Lastly, selecting the sensor module **150** allows a user to review the status (e.g., on/off) of the emergency LEDs **308**, and can also allow the user to turn these LEDs

on or off (or to adjust their intensities).

(78) The alert indicator **190** preferably displays alerts determined by the system even without selection by the user. In the depicted example, an alert has issued because the sensor module **150** has sensed an excess of carbon monoxide. Displayed alerts may be selectable to allow the user to review further details information (such as the exact level of carbon monoxide detected).

(79) The application **230** preferably also includes options to allows various programs to be set, named, and if necessary scheduled for use at particular times. Definition of a program can include input of any of the control signals necessary at any of the fixtures **10i**, including fan speed (Cai), UV LED chip **82** intensity (Cbi), and white LED chip intensity (Cci and Cdi). These control signals are shown in conjunction with a manual control option, whereby any of these control signals can be individually changed by the user at any time, thus overriding the current programming of the system. Lastly, the application **230** can include an option to allow for automated control of the system via the control algorithm **320**. As discussed above, such automatic control occurs in accordance with one or more environmental conditions that are sensed by the sensors **330** in the sensor module. One skilled in the art will realize that there are many different ways in which information may be organized for input to, or output from, the GUI in the application **230**, and FIG. **12** is therefore merely one example.

(80) To this point, it has been assumed that the fixtures **10** provide both illumination (e.g., using white LED chips **28**) and sterilization (e.g., using UV LED chips **82**). However, this is not strictly necessary. As shown in FIG. **13**, a fixture **10** may only comprise hardware components necessary to provide sterilization (e.g., the UV LED chips **82**, the fan **20**), etc. That is, fixture **10** may not necessarily provide for illumination. This is still beneficial, because ceiling-mountable fixtures like that shown in FIG. **13** can still provide UV sterilization to the environment in which the fixture is placed. A fixture of this sort may be particularly useful if the environment in question already has its own illumination sources.

(81) Many modifications to the disclosed fixture **10** can be made, and the fixture **10** can be used in different environments to useful ends. For example, the white LED chips **28** may not include significant peaks at either or both of 405 nm or 470 nm, although the inclusion of these wavelengths is preferred to further aid sterilization that the fixture **10** provides. In fact, the white LED chips **28** may not be used, and instead other white light sources (e.g., bulbs) could be used in the fixture **10**, with disinfection occurring strictly through use of the fan **20** and the UV sterilization box **14**. The UV sterilization box **14** could include UV radiation sources other than UV LED chips **82**. For example, various UV emitting bulbs could be used inside the UV sterilization box **14**.

(82) The fixture **10** and system can be used in environments where pathogens may be present, and in particular air borne pathogens. This can include hospitals, nursing homes, operating rooms, restrooms, kitchens, etc. Fixture **10** can also be used in a grow farm setting, in which light fixtures **10** are used to grow plants. For example, the disclosed fixture can be used in the context of the above-incorporated '900 patent, and can include the various improvements to a light fixture that are disclosed in that document.

(83) The disclosed fixture **10**, and/or the sensor module **150**, may be made portable, as disclosed in "Portable Disinfection Unit," by inventors John C. Higgins, Cody Renwick, Tim Gray, and Chris Fall, a patent application which is filed concurrently herewith [Ser. No. 17/317,669], and which is incorporated herein by reference in its entirety.

(84) Although particular embodiments of the present invention have been shown and described, it should be understood that the above discussion is not intended to limit the present invention to these embodiments. It will be obvious to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present invention. Thus, the present invention is intended to cover alternatives, modifications, and equivalents that may fall within the spirit and scope of the present invention as defined by the claims.

Claims

1. A system positionable in an environment, comprising: a sensor module comprising a plurality of sensors each configured to sense different environmental conditions in the environment, and first telemetry circuitry configured to wirelessly transmit the sensed environmental conditions; and a master fixture comprising second telemetry circuitry configured to wirelessly receive the transmitted sensed environmental conditions; a fan configured to draw air from the environment into the fixture; and a sterilization box configured to receive the drawn air and to output sterilized air for inclusion back into the environment, wherein the sterilization box comprises a plurality of ultra violet (UV) radiation sources configured to irradiate the drawn air with UV radiation along at least one path to produce the sterilized air; illumination sources configured to provide illumination to the environment, wherein the illumination sources emit light at a peak wavelength of approximately 405 nm and a second next-highest peak wavelength of approximately 470 nm to simultaneously disinfect the environment and provide human-safe lighting, wherein the illumination sources are distinct from the UV radiation sources; and controller circuitry configured with a control algorithm to automatically control the fan and the UV radiation sources in accordance with the sensed environment conditions.
2. The system of claim 1, wherein automatically controlling the fan and the UV radiation sources comprises simultaneously adjusting a speed of the fan and adjusting an intensity of the UV radiation sources.
3. The system of claim 1, wherein the control algorithm is configured to automatically and simultaneously control the fan, the UV radiation sources, and the illumination sources in accordance with the sensed environment conditions.
4. The system of claim 3, wherein automatically controlling the fan, the UV radiation sources, and the illumination sources comprises one or more of: turning on or off the fan; adjusting a speed of the fan; turning on or off the UV radiation sources; or adjusting an intensity of the UV radiation sources; turning on or off the illumination sources; or adjusting an intensity of the illumination sources.
5. The system of claim 1, wherein the fixture comprises: at least two holes for outputting sterilized air, an air flow path from the fan to each hole, and wherein at least one air flow path is non-linear.
6. The system of claim 1, wherein the UV radiation sources comprise UV LED chips configured to produce the UV radiation with a peak wavelength in the range from 200 to 280 nm.
7. The system of claim 1, wherein the sensor module and the master fixture are configured to be affixed to a ceiling in the environment.
8. The system of claim 1, wherein the sensor module further comprises a battery.
9. The system of claim 8, wherein the sensor module further comprises white LEDs to provide illumination to the environment, wherein the white LEDs are configured to receive power from the battery.
10. The system of claim 1, further comprising an alert indicator, wherein the control algorithm is configured to issue an alert to be broadcast by the alert indicator depending on the sensed environment conditions.
11. The system of claim 1, further comprising an application configured to be executed on an external device in wireless communication with the master fixture, and to generate a graphical user interface on the external device.
12. The system of claim 11, wherein the application is configured to receive at the graphical user interface a user input to initiate the control algorithm to automatically control the fan and the UV radiation sources in accordance with the sensed environment conditions.
13. The system of claim 12, wherein the application is further configured to receive at the graphical user interface one or more inputs to cause the control algorithm to manually control the fan and the

UV radiation sources in accordance with the one or more inputs.

14. The system of claim 12, wherein the application is further configured to receive at the graphical user interface a user selection of a program to cause the control algorithm to control the fan and the UV radiation sources in accordance with the selected program.

15. The system of claim 1, further comprising one or more secondary fixtures identical in construction to the master fixture but differently programmed from the master fixture, each secondary fixture also comprising: second telemetry circuitry; a fan configured to draw air from the environment into the fixture; and a sterilization box configured to receive the drawn air and to output sterilized air for inclusion back into the environment, wherein the sterilization box comprises a plurality of ultra violet (UV) radiation sources configured to irradiate the drawn air with UV radiation along at least one path to produce the sterilized air.

16. The system of claim 15, wherein the control algorithm in the master fixture is configured to automatically control the fan and the UV radiation sources in the master fixture and the one or more secondary fixtures in accordance with the sensed environment conditions.

17. The system of claim 16, wherein the second telemetry circuitry in the master fixture is further configured to wirelessly transmit one or more control signals to each of the secondary fixtures to automatically control the fan and the UV radiation sources in the one or more secondary fixtures.

18. The system of claim 17, wherein the second telemetry circuitry in each of the one more secondary fixtures is configured to receive the one or more control signals.

19. The system of claim 15, wherein the primary fixture and the one or more secondary fixtures further comprise illumination sources configured to provide illumination to the environment, wherein the control algorithm is configured to automatically control the fan, the UV radiation sources, and the illumination sources in the primary fixture and each of the one or more secondary fixtures in accordance with the sensed environment conditions.

20. The system of claim 1, wherein the fan draws air simultaneously into at least two different air flow paths within the fixture.

21. The system of claim 1, wherein the fan is a single fan that draws air simultaneously into at least four different air flow paths within the fixture.

22. The system of claim 1, wherein one or more of the at least one path comprises a portion of the path comprising an entry and exit, the entry and exit being the same, the portion defined by three or more baffles, the portion comprising a plurality of UV radiation sources, and the portion creating a vortice in the air flow path.
